OSF Amendment 01 — Two-measure framework

title: OSF Amendment 01 — Two-measure framework (epigraphic acts and epigraphic content)
amendment-number: 01
status: DRAFT for Shawn's review and lodgement (not yet lodged)
date-drafted: 2026-05-29
date-updated: 2026-06-04 (encoded the field-standard convergence gate — benign-divergence-tolerant, $\hat{R}$/ESS-based, replacing the zero-tolerance gate; re-scored Grid A to B = A = 98.6 %, the flat-null gap resolved as verified-benign divergences; diagnostic → Bland–Altman limits of agreement, shape-conditioned; checked against published Bayesian-workflow practice; clarified the §A8 repository-state provenance line — the git lodgement tag is the reproducibility anchor, distinct from the OSF-assigned identifier). Prior: 2026-06-03 (finalised §A5.5.1 + added §A5.7 against the completed adjudication); 2026-06-02 (added §A5.5.1).
scope: Two-measure framework (acts vs content) + recovery-grid binding-criterion clarification finalised against the completed two-grid adjudication (§A5.5.1) + subset-specific deconvolution and the measured reachability floor (§A5.7, Decision 34)
preregistration: https://osf.io/uycs6/ (lodged 2026-05-20; embargoed)
lodged-version: git tag osf-lodgement-2026-05-20 (https://github.com/saross/inscriptions/tree/osf-lodgement-2026-05-20)
filed-under: preregistration §7 / contingency clause (preregistration-draft.md line 423): substantive methodology changes after lodgement are filed as an OSF amendment before implementation.
author: Shawn Ross (with Claude Code as analyst/RSE)
gate: BINDING — Stage 3 confirmatory fits under the two-measure framework must not run until this amendment is lodged.

OSF Amendment 01 — Two-measure framework (epigraphic acts and epigraphic content)

Plain-language summary

Roman inscriptions are mostly dated by convention ("1st century AD") rather than to a year — a kind of dating **fog**. This project builds and validates a statistical tool that **de-fogs** a corpus: it recovers the real timeline of inscribing activity from the foggy dates, with honest uncertainty. In plain terms, this amendment does four things:

1. **Two measures, not one.** We count epigraphic activity two ways — number of *inscriptions* (acts of inscribing) and amount of *text* (letter mass) — and treat them as measuring *different things* (how often vs how much), reported side by side rather than one replacing the other.
2. **We say exactly when the de-fogging tool can be trusted.** Tested on simulated data with known answers, it works reliably for **inscription counting** but **not for letter counting** (the letter totals are too lopsided for the computation to behave). It needs a reasonably large group (roughly **2,000+** inscriptions in hard cases, fewer in easy ones) and breaks down once more than about **70 %** of the dating is fog. These boundaries are reported as the tool's "spec sheet."
3. **We corrected our own grading.** A first quality check was stricter than any standard statistical practice — it failed test cases for tiny, *harmless* computational hiccups. We verified the hiccups were harmless and relaxed the rule to the field standard, and the tool's validated pass rate rose accordingly (Grid A to 99 %). We also stopped treating the "fog fraction" as a precise number: it is only roughly recoverable, so we report it as a bounded, shape-dependent estimate used for coarse, directional statements only.

4. **Subsets are de-fogged on their own terms.** Applied to a subset (a province, an inscription category), the tool learns *that subset's* own cataloguing conventions rather than assuming the empire-wide pattern.

Every threshold and rule above is justified against published statistical practice (cited in §A5.5.1 and the provenance), so the choices rest on field standards, not our preferences. The technical statement follows.

A1. Identification and trigger

This is the first amendment to the project's preregistration (Open Science Framework, OSF, record `osf.io/uycs6/`, lodged 2026-05-20, currently embargoed). It is filed under the preregistration's own contingency rule (`preregistration-draft.md` line 423): *"If substantive methodology changes are required after lodgement … an OSF amendment is filed before implementation."*

Trigger. The 2026-05-26 letter-count probe (`runs/2026-05-26-letter-count-probe/REPORT.md`) and the methodological reframe it prompted, recorded in the project working notes (`docs/notes/reflections/working-notes.md`; commits `dd326dc`, `de8fa8f`, `2f86c95`).

Second trigger (2026-06-02; §A5.5.1). Adjudicating Grid A of the two-unit recovery simulation exposed that the lodged recovery-grid binding criterion is mathematically undefined for the flat genuine shape and gates on exact credible-interval coverage that collapses at large N for asymptotic reasons unrelated to recovery. The metric correction is folded into this amendment as §A5.5.1 rather than filed separately, since both changes concern the same recovery-grid / Stage-3 gate.

Third trigger (2026-06-03; §A5.5.1 finalisation + §A5.7). Both recovery grids are now complete (Grid A 2026-05-27; Grid B 2026-06-03) and have been adjudicated under the corrected criterion: **Grid A (inscription) PASSes; Grid B (letter) FAILs** — no letter-mass cell clears the convergence precondition. §A5.5.1 is finalised here with those results. Separately, a strategic decision on where the method's utility lies (Decision 34) pins **how a subset SPA is de-fogged** — by a *subset-specific* mixture fit, not by imposing the empire-wide convention shape — and a small-N reachability study (2026-06-03) measured the minimum subset size at which that subset-specific deconvolution recovers reliably. Both are pre-specified here as **§A5.7**, since they govern the same Stage-3 H3b mechanism this amendment otherwise leaves under-specified.

A2. Summary of the change

The lodged preregistration treats **letter count** as a single §5 pre-specified *exploratory* cross-check, reported alongside the inscription-count analyses but explicitly *"as a cross-check on the inscription-count results rather than a replacement for them"* (`preregistration-draft.md` line 388).

This amendment replaces that framing with a **two-measure framework**:

- **Inscription count** measures **epigraphic acts** (how often a community inscribed). It remains the project's **primary-by-convention** measure: it is the unit used by Hanson (2021), keeps the analysis comparable to the field, and is unchanged by this amendment.
- **Letter mass** measures **epigraphic content** (how much was inscribed). It is **promoted from an exploratory cross-check to a co-registered parallel confirmatory measure** of a *different construct* — neither a rival operationalisation of the same construct, nor a replacement for inscription count.
- The **delta between the two measures** (the content surplus or deficit per act) becomes a **pre-specified derived quantity**, reported descriptively as a second residual axis alongside the existing scaling residual.

The two measures are analysed **in parallel**, each within its own confirmatory family, and reported **side by side**. No multiplicity correction is applied *across* the two units, for the reason given in §A5.3.

This amendment additionally **clarifies the recovery-grid binding criterion** (§A5.5.1, a metric correction surfaced when Grid A was adjudicated on 2026-06-02): the lodged criterion is undefined for the flat genuine shape (Pearson r on a constant target) and gates on exact credible-interval coverage, which collapses at large N for asymptotic reasons unrelated to recovery. The corrected criterion patches the flat case with Wasserstein-1, keeps Pearson r 0.95 for all other shapes, demotes to a quantified diagnostic, and reports the grid as a recoverability map with a stated operating envelope. **Both grids are now adjudicated under the corrected criterion (§A5.5.1): inscription mass PASSes (the gate is met for the inscription unit); letter mass FAILs (its fits do not converge), so Stage-3 mixture-deconvolution work proceeds under inscription mass only.**

Finally, this amendment **pre-specifies how a subset SPA is de-fogged** (§A5.7, Decision 34): by a **subset-specific** mixture fit — the model learns the subset's own convention mix from the subset's own data — **not** by imposing the empire-wide convention shape on every subset. A reachability study measured the minimum subset size at which this recovers reliably, defining the operating boundary for the paper's core subset re-application case.

A3. Rationale

Inscription count and letter mass are **not two ways of measuring the same thing**. A flat inscription count treats a long monumental dedication and a three-word funerary fragment as equivalent units; letter mass registers the quantity of information each carried. The lodged preregistration already states this disagreement with Hanson (2021), who identified total lettering as the methodologically desirable measure but rejected it as impractical for fragmentary material (Hanson 2021, p. 142).

What the lodged preregistration did **not** do is follow the disagreement to its conclusion. If the two measures track partially different constructs — *acts* versus *content* — then asking "which is the better unit?" is the wrong question. Where they diverge is itself the finding, not a tie-breaker: terse frontier-military epigraphy deflates under letter mass; monumental and commercial epigraphy amplifies; high-count, low-content corpora (e.g. graffiti-rich assemblages) separate the two measures sharply. The project therefore adopts both measures and treats their **difference** as a third variable capturing inscription *style* and *function* that neither measure alone encodes. This is a larger and more defensible methodological contribution than substituting one unit for another.

A4. Relationship to already-observed exploratory results (transparency)

We record explicitly what had been observed at the time this amendment was conceived, so that the promotion of letter mass to confirmatory status can be judged on its merits.

The 2026-05-26 probe was an **exploratory** descriptive comparison run on LIRE v3.0 (it pre-dates and does not use the preregistered confirmatory pipeline). It found material divergence between the two units — a Hanson-style negative-binomial scaling exponent whose 95 % confidence intervals did not overlap between units, and substantial city- and province-level rank reshuffling (specifics in `runs/2026-05-26-letter-count-probe/REPORT.md` and the tables it cites). A subsequent Bayesian within–between (Mundlak) refit found the within-province population-attributable variance fraction `f_within` higher under letter mass.

This amendment is construct-driven, not result-driven, on three checks that a reviewer can verify:

1. **The framework refuses to pick a winner.** It does not adopt the measure that "did better"; it retains inscription count as primary-by-convention and adds letter mass as a parallel measure of a distinct construct. The divergence the probe found is treated as corroboration of the construct distinction, not as a reason to prefer one unit.
2. **The confirmatory decision rules for the letter-mass analyses are identical to the pre-existing inscription-count rules** (the same three-way `f_within` verdict; the same recovery-grid validation gate). No rule is tuned to the observed letter-mass result.
3. **Letter-mass Stage 3 fits remain gated** on the letter-mass recovery-grid simulation passing the same binding criteria as inscription mass (§A5.5). The amendment does not licence reporting any letter-mass confirmatory result that has not first cleared validation.

A5. Pre-specifications

A5.1 Measure definitions

- **Inscription count (acts).** Unchanged from the lodged preregistration: the per-subset, date-window-filtered inscription count specified in §3 and Decision 22.
- **Letter mass (content).** Summed `clean_text_conservative` characters (Latin A–Z only; Greek excluded), per the existing definition at `preregistration-draft.md` line 388. `clean_text_interpretive_word` is retained as a sensitivity variant. **Scope limitation, stated plainly:** because Greek is excluded, the content measure is *Latin* content; subsets and regions with substantial Greek epigraphy are under-counted on the content axis, and this is reported as a limitation wherever the content measure is used.

A5.2 Confirmatory structure under two units (scope)

The **letter-mass confirmatory family is deliberately bounded** to the cross-sectional analyses, to avoid requiring a new unit-specific power (reachability) simulation (see §A5.5 note):

- **Confirmatory under letter mass:** the **H3a within–between negative-binomial regression** (the three-way `f_within` verdict of `preregistration-draft.md` line 397) and the **H3a variance partition**.
- **Exploratory under letter mass (reported under both units, not confirmatory):** H3c spatial-residual analyses, H3b deviation-detection, and all §5 exploratories. Their confirmatory eligibility depends on detection-power thresholds (Phase 1; §6, lines 408–410), which are calibrated for an equal-weight *count* process and do **not** transfer to letter mass — a *compound sum* of heavy-tailed per-inscription letter counts. Empirically (`scripts/letter-mass-design-effect.py`), the letter-weight Kish design effect is large: corpus-wide ~15, per-city median ~2.4 (interquartile range ~1.9–3.7, at the `urban_context_city` analysis unit). The letter-mass detection SPA therefore carries *fewer* effective observations than the inscription-count SPA for the same inscriptions (0.42× at the median city), making letter-mass temporal detection *less* powered, not more. An analytic reachability translation (`scripts/letter-mass-reachability.py`) shows the consequence is categorical: **no** city in the corpus clears the preregistered urban-area detection thresholds under letter mass (0 of 1,044 Rome-excluded urban-area cities, versus 5–7 under inscription count) — even Pompeii, Salona, and Ostia fall below the floor once weighted by content. Letter-mass temporal detection is therefore not merely under-powered but unreachable corpus-wide; it cannot be confirmatory and is reported descriptively. Letter-mass time-series and residual analyses are consequently exploratory in this paper.

Inscription count retains its full pre-existing confirmatory family unchanged.

A5.3 Multiplicity

Each unit forms its own confirmatory family. No Holm–Bonferroni (or other) multiplicity correction is applied across the two units.

Justification: multiplicity correction guards against inflated false-positive rates when the *same* hypothesis is tested multiple ways. Here the two units operationalise **different constructs** (acts versus content) and therefore answer **different research questions**: "does within-province population explain variation in epigraphic *acts*?" and "…in epigraphic *content*?" are not two tests of one hypothesis. Correcting across them would penalise the project for asking two questions rather than one, and would misrepresent the inferential structure. Within each unit, the existing multiplicity policy is unchanged. The H3a verdict is rendered **separately and explicitly per unit**; the paper does not combine them into a single verdict.

A5.4 The inter-measure delta (derived, exploratory)

A **pre-specified derived quantity**, the **content residual**, is computed and reported descriptively:

- **Definition.** For the cross-sectional city set, fit `log(letter_mass) ~   log(inscription_count)` across cities; the per-city residual is the content residual (positive = more content per act than the corpus norm; negative = less). The project thereby has a **two-dimensional residual space**: scaling residual (observed inscriptions minus what population predicts) × content residual (observed content minus what inscription count predicts).
- **Status.** Exploratory and descriptive. **No pre-committed threshold and no confirmatory verdict** attach to the delta; it is a novel quantity and is reported as a map/descriptive characterisation and cross-tabulated against the scaling residual. It does **not** trigger additional preregistration amendments.
- **Subsumption.** The previously logged standalone "cumulative-totals Hanson negative-binomial experiment (inscription count and letter count)" (the project's tertiary backlog item 5) is **subsumed** into this two-measure framework rather than run as a separate exploratory.

A5.5 Recovery-grid validation under two units (the gate)

The lodged preregistration's H2.1 recovery-simulation gate (`preregistration-draft.md` §3 line 61; §4 line 334; §6 lines 395–396) — **(i)**   90 % of grid cells achieve per-cell   coverage and **(ii)** posterior-median Pearson r 0.95 between recovered and true genuine SPA in   90 % of cells — now runs as **two parallel grids**, one per unit:

- **Grid A — inscription mass** and **Grid B — letter-mass conservative**, under the same corrected (F1+F3, empirical-Bayes) pipeline, at `runs/2026-05-26-recovery-grid-two-unit/` (spec at `…/spec.md`). **Both grids**

are complete (Grid A finished 2026-05-27; Grid B 2026-06-03) and are adjudicated under the corrected criterion in §A5.5.1.

- **Gate rule.** A Stage 3 confirmatory fit that **uses the temporal mixture deconvolution** under a given unit is permitted only if **that unit's grid passes the binding criterion** (§A5.5.1). The outcome branching is pinned in `…/spec.md` §5: both grids pass → both units; one passes → that unit only; both fail → the mixture is revised before any Stage 3 fit (lodged recovery-failure contingency, line 420).
- **Realised outcome (2026-06-03): Grid A PASS / Grid B FAIL** (§A5.5.1) → the temporal mixture deconvolution is validated for **inscription mass only**. This is fully anticipated by §A5.2: the letter-mass *confirmatory* family was already bounded to the cross-sectional H3a (per-city totals regressed on population), which does **not** use the temporal mixture deconvolution; letter-mass temporal/detection analyses were already exploratory. Grid B's failure therefore **reinforces** the existing scope rather than changing it — no letter-mass temporal-deconvolution confirmatory was ever claimed, and the letter-mass H3a cross-sectional confirmatory is unaffected.
- **Note (why letter-mass confirmatory is scoped to the cross-section).** The recovery grid validates the **mixture model** (the temporal deconvolution) under each unit; it does **not** establish **detection-power** thresholds, which are a separate Phase-1 product and cannot be re-used for letter mass. Letter mass is a compound sum of heavy-tailed per-inscription letter counts (per-city Kish design effect median ~2.4; §A5.2), so its temporal-detection power is design-effect-limited and *lower* than inscription count's. The cross-sectional H3a regresses per-city letter-mass *totals* on population and does not use detection thresholds (see line 131), so the design effect does not bear on it — confining the letter-mass *confirmatory* family to the cross-section is therefore principled, not a convenience. An analytic reachability translation (`scripts/letter-mass-reachability.py`) already shows letter-mass temporal detection is unreachable for every urban-area city in the corpus (0 of 1,044, versus 5–7 under inscription count); because the neglected heavy-tail effects can only reduce power further, a full compound-process simulation cannot overturn this. The simulation is therefore logged as an optional methodology-follow-up refinement, not a prerequisite; the paper reports the analytic reachability result.

A5.5.1 Binding-criterion clarification (metric correction)

Adjudicating Grid A (inscription mass) on completion (2026-06-02; `runs/2026-05-26-recovery-grid-two-unit/inscription-mass/outputs/REPORT.md`) exposed two defects in the lodged binding criterion as written — both *mathematical / asymptotic*, not recovery failures. This subsection corrects them. The correction was checked against field practice (`planning/prior-art-scout-2026-06-02-recovery-validation-metrics.md`).

The two defects.

1. **Criterion (ii) is undefined for the flat genuine shape.** Pearson r between a recovered curve and a *constant* true SPA is `0/0` (the truth has zero variance). All 75 `flat_baseline` cells return `nan` and fail mechanically, capping achievable shape-pass at 83.3% — independent of model quality, and identically for both units. The model in fact recovers flatness well (99% cell coverage; small Wasserstein-1).
2. **Criterion (i) — exact 95%-CI coverage of — collapses at large *N*.** Holding (shape, , tier) fixed and increasing N, cell coverage falls from ~1.0 to ~0.0 while the bias stays small and roughly constant — the standard posterior-concentration / semiparametric Bernstein–von Mises effect (the interval narrows below a fixed small bias). It measures asymptotic interval calibration, not recovery adequacy. No surveyed community (radiocarbon SPD, baorista, Bayesian-workflow SBC) gates on exact CI coverage of a mixing parameter.

The corrected criterion. The recovery grid is reported as a **recoverability map** with a stated **operating envelope** (consistent with the project's reachability-guide methodology), not a binary whole-grid pass/fail. Within the operating envelope:

- **Precondition — convergence.** A cell is eligible only if 90 % of its replicates meet **R < 1.01 and bulk-ESS 400** (the lodged convergence thresholds; Vehtari et al. 2021). **Divergences are handled per field standard** (Stan diagnostics guidance; Betancourt 2017), *not* as a per-replicate zero-tolerance auto-fail: a few scattered post-warmup divergences are investigated, not automatically rejected. A cell whose replicates carry divergences remains eligible if those divergences are **benign** — low-rate and not degrading recovery (the recovery of diverging replicates is statistically indistinguishable from non-diverging ones) — and is failed only if they are **clustered or step-size-persistent** (recovery-degrading). This replaces an earlier zero-tolerance divergence gate (`n_divergences == 0` per replicate), which we confirmed (2026-06-04) is stricter than any field source endorses (no surveyed authority specifies a divergence-rate threshold; the standard is contextual investigation). On Grid A the only divergent cells are the flat null, and their divergences are verified benign (§"Flat-null divergences are benign", below), so they pass the precondition.

- **Per-cell shape gate — genuine-SPA (`p_gen`) shape recovery (hybrid).** posterior-median Pearson r 0.95 for **non-flat** genuine shapes (**unchanged from the lodged preregistration**); and, for the **flat_baseline** shape only (where Pearson r is undefined), Wasserstein-1 between posterior-median recovered and true SPA **T_flat = 10 years** (the maximum W1 among well-recovered flat cells is 9.8 y; rounded up). Wasserstein-1 is additionally reported for *all* shapes as a supplementary distribution-sensitive metric. A single global W1 threshold is *not* used, because W1 magnitude depends on the true shape's geometry (good-recovery W1 ranges 0.8–24 across shapes) and would penalise high-spread shapes; only the *undefined* flat case is patched.
- **Grid-level pass rule (headline B).** A unit's grid **passes** if **90 % of all in-envelope cells are clean passes** — i.e. they both meet the convergence precondition AND pass the shape gate. Counting non-converging cells as failures (rather than dropping them) is the conservative choice and keeps any genuine convergence pathology visible in the headline. A **diagnostic figure (A)** — the shape-pass rate among the convergence-*eligible* in-envelope cells only — is reported alongside. Under the field-standard convergence gate above, B and A **coincide** on Grid A (no benign-divergence exclusions remain); they would diverge only for a unit with genuinely non-converging (R/ESS-failing) in-envelope cells.
- **Operating envelope.** The binding criterion is evaluated where the deconvolution is identifiable: empirically **0.70**. A finer- run (2026-06-04; {0.70 … 0.90} at N = 2,000; `runs/2026-06-04-envelope-finer-alpha/`) shows recovery declines **gradually and shape-dependently** above the cut rather than at a sharp cliff — mean shape-recovery 83 % (= 0.70) → 67 % (0.75) → 44 % (0.80) → 17 % (0.90), with simple trends (smooth_growth) holding to 0.80 and multimodal shapes (regnal_cluster) collapsing by 0.75–0.80. So 0.70 is the last broadly-reliable region (90 % in the full grid's shape/tier/N mix), and recovery within (0.50, 0.70) is already shape-dependent and is read off the recoverability map per shape. Cells with 0.95 (5 % genuine signal; near-unidentifiable) are reported as a **stress-test sensitivity, not gated**. Where the *real* corpus exceeds the envelope (plausible in late, template-dominated sub-periods), genuine-signal claims for those regimes are flagged as degraded-recovery, not validated.
- **(mixing weight) — diagnostic, not gate.** recovery is reported as a **quantified diagnostic**, not a binding gate. Rationale: (a) exact CI coverage of a mixing weight is not field-standard and collapses at large N under negligible bias; (b) the recovery error is *structural* — it does **not** shrink with N. Following method-comparison practice (Bland–Altman limits of agreement; Kruschke ROPE), recovery precision is reported as **95 % limits of agreement [−0.22, +0.17]** (mean signed bias −0.02 — a slight under-estimate), and is **conditioned on the data-generating shape**, since it varies 4× by shape complexity: 90th-percentile |bias| **0.07–0.11** for smooth/flat shapes, rising to **0.18–0.27** for multimodal shapes (bimodal, regnal_cluster). **All -derived claims in the paper are hedged to this demonstrated, shape-conditioned recovery precision** — coarse directional convention-fraction statements are supported; a tight dial is not.

Finalised adjudication (2026-06-03; transparency). Both grids' stored outputs were re-scored under the corrected criterion (no re-fitting — W1, intervals, and convergence are stored per cell). Each grid carries 450 cells (3 tiers × 5 × 6 shapes × 5 N); the operating envelope (0.70) holds 360.

- **Grid A — inscription mass: PASS.** Headline **B = 98.6 %** (355/360 in-envelope cells are clean passes), which **equals diagnostic A**: under the field-standard convergence gate all 360 in-envelope cells are eligible, and the only 5 failures are non-flat cells missing the Pearson gate at = 0.70. Under the *lodged* criterion the same grid (whole-grid both-pass) returned only 42.7 %; the difference is the §A5.5.1 metric correction and operating-envelope reframe re-scoring the **same** stored fits — not any change to the fits.
- **Grid B — letter mass: FAIL.** Headline **B = 0.0 %**: **no** in-envelope cell reaches the 90 % convergence precondition — and crucially this holds **even under the field-standard gate** (dropping the divergence requirement recovers **zero** letter cells), so letter mass fails on **R/ESS**, not merely on divergences. It is a genuine sampling failure: letter mass is a *compound sum* of heavy-tailed per-inscription letter counts (divergences run to ~13,000 per cell; median recovered Pearson r 0.76) whose likelihood geometry the NUTS sampler cannot navigate reliably. It empirically confirms the analytic finding that letter-mass temporal detection is unreachable corpus-wide (§A5.2), and is consistent with inscription count being the primary unit of analysis.

Outcome branch: PASS / FAIL → the temporal mixture deconvolution proceeds under **inscription mass only**; letter mass is reported as a documented limitation (its cross-sectional H3a confirmatory does not use the mixture and is unaffected — §A5.5).

Flat-null divergences are benign (resolved 2026-06-04, not a limitation). An earlier draft reported a B A gap on Grid A from 24 flat-null cells failing a zero-tolerance divergence gate. That gate is replaced by the field-standard convergence treatment above, and the underlying question is settled empirically. Across the 6,000 in-envelope flat replicates, 10.2 % carry 1 divergence (median 1; maximum rate 0.36 % of draws); these diverging replicates recover the flat shape **no worse** than the non-diverging ones (median W1 0.59 vs 0.55; **99.7 %** vs 97.8 % within the W1 10 y gate; Mann–Whitney p 0.36) and all pass $\hat{R}$/ESS. By the Stan/Betancourt spatial-and-recovery criterion these are scattered false-positive divergences, not pathological ones, so the cells are eligible and Grid A's B and A coincide at **98.6 %**. (Mechanism: under a flat genuine SPA the GRW genuine-signal latent is near-unidentified even after non-centred reparameterisation, producing occasional benign divergences; documented as a known, harmless model property — no re-fit is required.)

Metric-correction-driven, not verdict-driven (the integrity checks a reviewer can verify). Mirroring §A4:

1. The correction targets two *mathematical / asymptotic* defects (undefined-on- flat; coverage-collapse-at-large-N), identified from the failure structure independently of the verdict, and confirmed field-standard against published practice (artefacts above).
2. The non-flat shape criterion is **unchanged** from the lodged preregistration (Pearson r 0.95); only the *undefined* flat case is patched (W1), and is moved to a diagnostic.
3. Thresholds (T_flat = 10 y from well-recovered flat cells; operating envelope 0.70 from near-unidentifiability at 0.95) are set from theory and the known-good sub-grid **before** the headline two-grid verdict; the failing scenarios (full-grid; -gated) are reported alongside the passing one.

A5.6 Exploratory analyses under both units

The §5 pre-specified exploratory analyses are run under **both** units where applicable, with the per-subset delta reported as data. These remain exploratory (no confirmatory verdicts), exactly as in the lodged preregistration.

A5.7 Subset deconvolution — subset-specific fits and the reachability floor (Decision 34)

The lodged preregistration scans mixture-corrected SPAs on subsets (H3b) but does not pin **how** a subset SPA is corrected, and the Stage-3 convention component `p_conv` is estimated corpus-wide. The only specified path was therefore to impose the empire-average convention shape on every subset. This amendment pre-specifies the mechanism (Decision 34).

- **Subset SPAs are de-fogged by a subset-specific mixture fit.** The model (`build_model_f1_f3`) learns the subset's *own* convention mix (`tier_weights`, the editorial-template composition) from the subset's own data; only the universal template-width *basis* is fixed, never corpus content. **The empire-wide `p_conv` is NOT imposed on subsets.** A subset with its own convention profile (a Greek-East province, an epitaph-heavy subcategory) is fitted on its own terms. This supersedes the H3b "empire-correction-applied-to-subsets" reading; it is the methodological core of the paper's re-application case (a coherent subcorpus gets a de-fogged temporal trajectory with honest uncertainty, where before there was an artefact-contaminated histogram). The empire-scale fit remains a proof of concept and a candidate empire-level proxy, not the engine of subset analysis.
- **Feasibility is N-dependent; the floor is measured.** A standalone per-subset fit is harder to identify at small N. The small-N reachability study (`runs/2026-06-03-small-n-reachability/`; 4,200 fits over 84 cells = 3 shapes × 4 × 7 N, scored under the §A5.5.1 criterion) measured the minimum subset size N at which subset-specific deconvolution recovers reliably. **Within the operating envelope (0.70), reliable recovery has a worst-case floor of N 2,000**, falling to **N 500** for the easiest subsets (low convention fraction 0.30, smooth or rise-and-fall signals). Two = 0.70 cells remain unreached even at N = 2,000, and the = 0.85 stress row is unreached throughout. Mean shape-recovery rate (0.70) rises 12 % (N = 50) → 94 % (N = 2,000); convergence 100 %; mean | -bias| 0.13 (inside the ±0.18 envelope precision). This replaces the prereg's rough "unidentified below N 100" prior with a measured reachability map (`…/outputs/REPORT.md`, `…/outputs/figures/reachability-map.png`).
- **Below the floor.** A subset below its reachability floor is not de-fogged on its own; the fall-backs are partial-pooling of the convention across subsets (a §5-style borrow), the §5 hierarchical aoristic trajectory model, or descriptive reporting. These are out of scope for confirmatory H3b and reported as such.
- **Eligibility.** H3b subset deconvolutions are gated by **both** the measured reachability floor (this subsection) and the existing Phase-1 detection thresholds. The paper reports, per subset, whether it clears the floor.

A note on credible bands (limitation carried forward). The recovered `p_gen` credible band is honest for smooth timelines but **overconfident for sharply-peaked signals, and degrades with N**: in the band-calibration diagnostic (`runs/2026-06-02-recovery-utility-check/`) mean pointwise 95 % coverage falls from 0.90 at N = 2,000 to 0.67 at N = 50,000 (worst for `regnal_cluster`) — the same posterior-concentration mechanism that demoted , compounded by the Gaussian-random-walk smoothness prior being unable to represent sharp features. The **median** (point) timeline — the gated quantity — stays trustworthy; reported **bands** in peaked regimes are widened/caveated. The real corpus has sharp regnal clustering, so this is material and is reported as a limitation, not a new gate.

A6. What does NOT change

- Inscription count remains the **primary-by-convention headline measure**; comparability with Hanson (2021) is preserved.
- The inscription-count confirmatory family, its decision rules, and its multiplicity policy are **unchanged**.
- The chronological envelope (50 BC – AD 350), the LIRE v3.0 freeze (Decision 24), and all other lodged specifications are **unchanged**.
- The recovery-grid is applied per unit; the R/ESS convergence gates and PPC trigger scheme are **unchanged**. The recovery-grid **binding criteria are clarified** (metric correction) per §A5.5.1: the undefined-on-flat Pearson case is patched with Wasserstein-1, exact -coverage is replaced by an diagnostic, and the grid is reported as a recoverability map with a stated operating envelope. The non-flat shape criterion (Pearson r 0.95) is unchanged.

A7. Changes by preregistration section (a fresh upload, not in-place edits)

The lodged preregistration cannot be edited in place; this amendment is uploaded to OSF as a **standalone addition**. The list below keys the amendment's changes to the lodged preregistration's sections, for the reader's cross-reference — it is **not** a set of in-place edits to the original. (The project's living working copy is maintained separately; the lodged authority remains git tag `osf-lodgement-2026-05-20`.)

1. **§3 (H3a specification):** add letter mass as a co-registered parallel confirmatory measure for H3a + variance partition, with the per-unit separate-verdict and no-cross-unit-correction statement (§A5.2, §A5.3).
2. **§5 line 388 (letter-count alternative analysis):** reframe from "exploratory cross-check / lesser of two evils / not a replacement" to the two-measure framework; cross-reference the content-residual derived quantity.
3. **§5:** add the content-residual derived quantity (§A5.4); note §5 exploratories run under both units; note subsumption of backlog item 5.
4. **§6 effect-size table:** add (a) a letter-mass H3a `f_within` verdict row (same three-way rule), (b) a second H2.1 row for the letter-mass recovery grid, and (c) a content-residual descriptive row (no threshold).
5. **§3 line 61 / §4 line 334 / §6 lines 395–396 (recovery-grid binding criterion):** apply the §A5.5.1 metric correction — patch the flat-shape case with Wasserstein-1 (T_flat = 10 y), retain Pearson r 0.95 for non-flat shapes, make the convergence precondition explicit (R < 1.01 and bulk-ESS 400, with divergences handled benign-tolerantly per Stan/Betancourt — not a zero-tolerance auto-fail), demote exact -coverage to a shape-conditioned -recovery diagnostic (Bland–Altman limits of agreement), and reframe the gate as a recoverability map with a stated operating envelope (0.70). State the grid-level pass rule as **headline B** (90 % of all in-envelope cells are clean passes — converge AND pass the shape gate), with **diagnostic A** (shape-pass among convergence-eligible cells) reported alongside; record the realised outcome (**Grid A PASS, B = A = 98.6 % / Grid B FAIL → mixture deconvolution under inscription mass only**). Update both H2.1 rows in the §6 table accordingly.
6. **§3 / H3b mechanism (Decision 34; §A5.7):** specify that subset SPAs are de-fogged by a **subset-specific** mixture fit (the model learns the subset's own convention mix; the empire-wide `p_conv` is **not** imposed on subsets), superseding the "empire-correction-applied-to-subsets" reading; gate H3b subset deconvolutions on the **measured reachability floor** (§A5.7; worst-case N 2,000 within 0.70) in addition to the existing Phase-1 detection thresholds; note the below-floor fall-backs.
7. **§7 / §11:** record this amendment in the planned-deviations text and the §11 post-lodgement amendment trail.
8. **`preregistration-changelog.md`:** add a dated amendment section.

A8. Provenance

- **Statistical reasoning:** the load-bearing choices in this amendment are the no-cross-unit-correction decision (§A5.3); the scoping of the letter-mass confirmatory family to the cross-section (§A5.2, §A5.5); the §A5.5.1 metric correction with its headline-B / diagnostic-A reporting and flat-null treatment; and the subset-specific deconvolution mechanism (§A5.7, Decision 34). All are justified inline.
- **Artefacts:** `runs/2026-05-26-letter-count-probe/` (probe); `runs/2026-05-26-recovery-grid-two-unit/` (two-grid validation + the finalised cross-grid adjudication, `comparison/COMPARISON-REPORT.md`); `runs/2026-06-03-small-n-reachability/` (the reachability floor); `runs/2026-06-02-recovery-utility-check/` (band-calibration + real-corpus diagnostics); the project working notes (`docs/notes/reflections/working-notes.md`).
- **Binding-criterion clarification (§A5.5.1):** the metric correction is driven by the Grid A adjudication (`planning/prior-art-scout-2026-06-02-recovery-validation-metrics.md`). Field basis: no surveyed community gates on exact CI coverage of a mixing weight; Wasserstein-1 is theoretically justified for deconvolution recovery (Rousseau & Scricciolo 2021); flat/uniform is a standard tested null in radiocarbon SPD work (Crema 2022). **Finalised against the completed two-grid adjudication 2026-06-03** (harness updated to compute the corrected criterion alongside the lodged one, with a built-in regression check reproducing the Grid A figures B 91.9 % / A 98.5 % under the then-current zero-tolerance gate — superseded 2026-06-04 by the field-standard gate, re-scoring Grid A to B = A = 98.6 %; cross-grid adjudication committed `1bf791f`). The earlier flat-null limitation is **resolved** (divergences verified benign; the re-fit is retired); its history is logged at `planning/backlog-2026-05-03.md` (§Phase-2 refinements).
- **Subset deconvolution + reachability floor (§A5.7; Decision 34):** the subset-specific-fit decision and the measured floor are recorded in the decision log (Decision 34) and `runs/2026-06-03-small-n-reachability/` (4,200-fit study; floor + map committed `5601b04`); the paper-facing rationale is `planning/paper-significance-and-applications-2026-06-03.md`.
- **Repository state:** the amendment text and its supporting artefacts are committed on `main`; the lodged repository state is anchored by a new git tag, `osf-amendment-01-<lodgement-date>` (cut at lodgement, mirroring `osf-lodgement-2026-05-20`). This git tag is the project's own reproducibility anchor and is **distinct from** the OSF-assigned registration/amendment identifier, which OSF generates on approval.